EXECUTIVE VALUE PROPOSITION & SOLUTION BRIEF

Pandita's Aether-DCTWIN

Autonomous, Physics-Informed Digital Twin & Supervisory Control Platform for High-Density AI & Colocation Critical Facilities

Hybrid Air (CRAC) + Direct-to-Chip Liquid (CDU) | 100% Agentless Facility Demarcation | Online Self-Calibrating Physics | BACnet Priority-8 Safe Control | 16 Integrated Operational Modules

The AI Data Center Inflection Point: Why Legacy BMS & DCIM Fall Short

As enterprise and colocation data halls transition from **12–20 kW air-cooled racks** to **65–120 kW direct-to-chip liquid-cooled AI clusters**, Critical Facilities teams face an acute operational squeeze that reactive PID loops and static DCIM dashboards cannot solve:

1. THERMAL VELOCITY RISK

Seconds, Not Minutes, of Margin

Synchronized LLM training bursts ramp rack heat 4× faster than chilled-water thermal loops respond. Under a CRAC or CDU pump trip, cold-aisle and manifold SLA ride-through shrinks from 12 minutes to **under 110 seconds**.

2. STRANDED MULTI-MILLION CAPEX

15–25% Locked-Away Capacity

Static year-round over-cooling setpoints, unbalanced 3-phase (L1/L2/L3) row PDU whips, and slow spreadsheet capacity checks leave millions of dollars in sellable power and cooling stranded.

3. THE "BLACK-BOX AI" VETO

Security & Uptime Resistance

Colocation and enterprise operators strictly prohibit installing telemetry agents inside tenant servers—and Chief Engineers refuse unguarded AI controllers that risk condensation or SLA breaches.

Headline Customer Value Proposition (Reference 5 MW Hybrid AI Hall)

ENERGY & PUE CUT

8% – 18%

Drops PUE from `1.34` → `1.18` via Constrained MPC, VFD cubic affinity & wet-bulb free cooling.

RECURRING OPEX SAVED

$700K/yr

Per 5 MW hall (**$1.4M+/yr at 10 MW**) across HVAC kWh, utility TOU peak shaving & predictive CBM.

STRANDED CAPACITY

+13–20%

Reclaims ~650 kW per 5 MW hall—worth **$5.5M in avoided Tier-III CapEx** or **+$1.17M/yr** colo lease revenue.

TIME-TO-VALUE

< 5 Mos

Calibrates from existing BMS/EPMS in 48 hrs; zero server agents; zero-downtime 30-day Shadow Pilot.

The Pandita's Aether-DCTWIN Core Promise

Pandita's Aether-DCTWIN unifies an online self-calibrating 3D electro-thermal digital twin, mathematically constrained Model Predictive Control (`SLSQP + Control Barrier Functions`), Tier-III N–1 compliance auditing, colocation capacity fit-out simulation, and automated NOC workflows—operating **100% on the Critical Facilities side of the demarcation line**.

Built for Colocation & Wholesale Operators

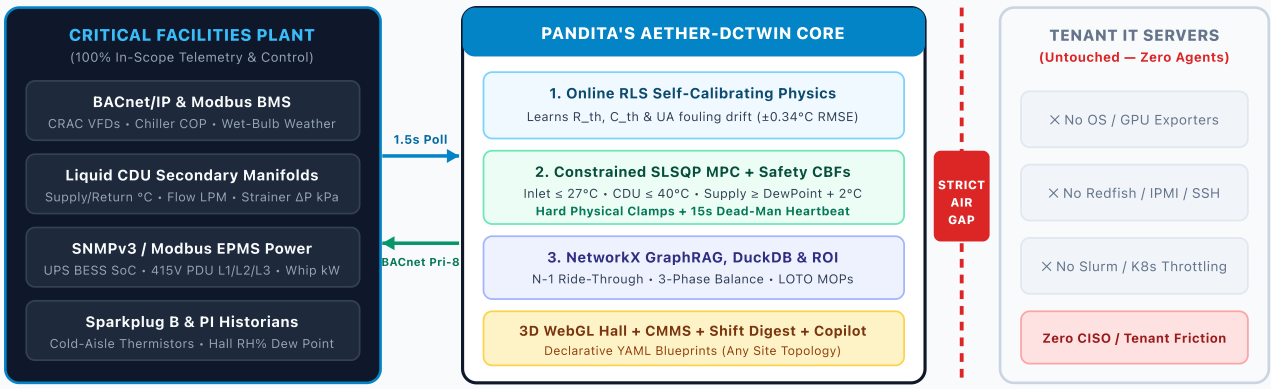
- Qualify prospect 500 kW+ high-density AI deployments in **60 seconds** instead of 2 weeks of engineering studies.
- Defend against disputed tenant SLA credits with cryptographic demarcation ledgers & automated CSRD/WUE certificates.

Built for Enterprise & Sovereign AI Factories

- Protect $100M+ GPU clusters from CDU strainer fouling, manifold condensation, and hot-aisle recirculation trips.
- Shave utility transformer peak-demand penalties using existing UPS Li-Ion BESS & chilled-water thermal inertia.

Why Operators Trust Us: "Church & State" Demarcation + BACnet Safety

Every capability in Aether-DCTWIN is engineered around the non-negotiable organizational boundary of real data centers: **zero software agents, zero SSH/IPMI credentials, and zero workload interference inside tenant IT servers.**



The 3-Stage Enterprise Trust Progression (Solving Day-1 Adoption)

No Chief Engineer enables closed-loop AI control on Day 1. Aether-DCTWIN builds operational confidence through a hardware-enforced 3-stage governance model:

STAGE 1 • DAYS 1–21

Shadow-Mode Pilot Validator

Runs **100% read-only** alongside existing BMS PID loops. Computes counterfactual kWh/$ savings and proves `0` ASHRAE thermal or dew-point breaches across real peak load swings before touching a single actuator.

STAGE 2 • DAYS 22–30

NOC Human-in-the-Loop

Presents recommended CRAC/CDU setpoint shifts in a 1-click NOC approval queue with predicted `ΔkW`, `ΔT`, and counterfactual sandbox preview—recording an immutable operator governance log.

STAGE 3 • PRODUCTION

BACnet Priority-8 Watchdog

Writes bounded supervisory setpoints to **BACnet Priority Array Slot 8** with a `15s` dead-man heartbeat. Any network loss or thermal excursion writes `NULL`—reverting instantly to local OEM Priority-16 PID.

Why Physics-Informed AI Beats Pure "CFD Consulting" & "Black-Box ML"

DIMENSION	STATIC CFD CONSULTING	GENERIC "BLACK-BOX" ML	PANDITA'S AETHER-DCTWIN
Calibration Speed	6–12 weeks manual CAD meshing; stale after first rack move	Requires months of fault data that operators don't have	**< 48 hrs** via declarative YAML + online RLS parameter learning
Unseen Fault Safety	Offline desktop tool; zero real-time protection	Extrapolates dangerously outside training distribution	**First-Law thermodynamics + hard SLSQP Control Barrier Functions**
Liquid + Air Coupling	Separate air vs. hydronic studies	Treats sensors as flat uncoupled features	**Coupled air CRAC + direct-liquid CDU + 3-phase PDU graph**

Complete 16-Capability Solution Architecture

Combining the core physics-informed Digital Twin engine with all 10 operator expansion modules delivers an end-to-end command center spanning **Uptime Assurance, Colocation Revenue Growth, Energy/Grid Arbitrage, and 24x7 NOC Automation**:

Pillar A — Tier-III Uptime Assurance, Physical Safety & Control Governance

Eliminates Outages, Pipe Sweating & Pilot Risk

1. Shadow-Mode Validator & BACnet Watchdog

GOVERNANCE

3-stage read-only shadow scoreboard, NOC approval queue, and BACnet Priority-8 dead-man relinquish to Priority 16.

2. Thermal Ride-Through & Tier-III N-1 Auditor

RESILIENCE

Continuously simulates single-CRAC/CDU trips every 30s, computing exact seconds-to-27°C runway vs. generator spin-up.

3. Psychrometric Dew-Point & Condensation Guard

LIQUID SAFETY

Magnus-Tetens hall dew-point tracking that clamps CDU coolant supply ≥ `DewPoint + 2.0°C` so manifolds never sweat.

4. GraphRAG Blast-Radius & LOTO MOP Generator

COMPLIANCE

Traverses NetworkX power/cooling topology to author ASHRAE TC 9.9 maintenance procedures with T-15m pre-cool steps.

Pillar B — Colocation Revenue Acceleration, Stranded Capacity & Tenant Trust

Unlocks Sellable MW & Defends SLAs

5. Colo Fit-Out & Summer Design-Day Planner

REVENUE

Validates prospect 250–1,000 kW AI deals in 60s under ASHRAE 99.6% summer wet-bulb + N-1 loss, issuing Deal Feasibility PDFs.

6. 3-Phase (L1/L2/L3) PDU Stranded kW Monitor

CAPEX UNLOCK

Computes 120° phasor neutral current & NEC 80% phase imbalance, prescribing exact rack whip swaps to reclaim stranded kW.

7. Tenant SLA "Blame-Defense" & ESG Certificates

COLO TRUST

Separates facility supply compliance from tenant over-draw disputes and exports monthly CSRD Scope-2, PUE & WUE statements.

8. 3D WebGL Hall, 42U Inspector & What-If Cloner

SPATIAL TWIN

Interactive 3D data hall with CFD exhaust plumes, liquid loop pulses, 42U rack drill-down & non-destructive sandbox cloning.

Pillar C — Autonomous PUE Optimization, Free Cooling & Utility Grid Arbitrage

Cuts Cooling kWh & Peak Demand

9. Constrained SLSQP MPC Cooling Controller

AI CONTROL

30-min receding-horizon solver that pre-cools ahead of AI bursts, enforces VFD slew limits & eliminates CRAC fan hunting.

10. Wet-Bulb Free-Cooling Economizer Advisor

THERMODYNAMICS

Detects live waterside economizer windows from outdoor wet-bulb & approach temps to maximize chiller-off free-cooling hours.

11. Utility TOU Tariff & UPS BESS Peak-Shaving

GRID FINOPS

Coordinates UPS Li-Ion BESS discharge and chilled-water thermal coasting during peak utility tariff windows & DR events.

12. Executive ROI, 5-Yr NPV & CapEx Engine

CFO REPORTING

Quantifies real-time dollar, MWh, tCO_2, water, and unlocked GPU capacity into board-ready financial business cases.

Pillar D — 24x7 NOC Productivity, Predictive CBM & Turnkey Integration

Slashes Alarm Fatigue & Emergency Truck Rolls

13. OpenDT Online RLS Self-Calibration

SELF-LEARNING

Stateful Kalman/RLS estimator tracking zone R_th & C_th online, cutting thermal error from ±3.0°C to ±0.34°C automatically.

14. Filter & CDU Strainer CBM RUL Tracker

MAINTENANCE

Models ΔP/flow fouling curves to predict Remaining Useful Life days and picks the ROI-optimal filter/strainer service date.

15. Causal Alarm De-Dup & CMMS Dispatch

NOC OPS

Collapses 50+ cascading rack threshold alerts into 1 upstream mechanical root cause and dispatches ServiceNow/Maximo payloads.

16. Shift Handover Digest, 24h DuckDB & Copilot

AUTOMATION

1-click 8h/12h shift sign-off reports, 24h Parquet incident scrubber, BYOD CSV fuzzy mapper & tool-calling AI engineering agent.

Quantified Financial Impact: Reference 5.0 MW Hybrid AI Data Hall

Below is a conservative, engineering-backed financial model for a single **5.0 MW IT load** hybrid data hall (`40% Perimeter Air + 60% Direct-to-Chip Liquid AI`, blended tariff `$0.115/kWh`, peak demand charge `$18.50/kW-mo`). Multi-hall campuses (`20–50 MW`) scale linearly:

VALUE DRIVER / CAPABILITY LEVER	PHYSICAL & OPERATIONAL MECHANISM	ENERGY / KW IMPACT	ANNUAL VALUE (USD)
1. SLSQP MPC & VFD Cubic Affinity (Modules #1, #9, #13)	Raises CRAC/CDU supply setpoints within ASHRAE A1/W2 envelopes; trims fan/pump VFD speeds ($P \propto Speed^3$) & stops hunting.	-310 kW avg 2,715 MWh/yr	**$312,200 / yr**
2. Wet-Bulb Waterside Economizer (Modules #3, #10)	Captures missed shoulder-season & overnight free-cooling windows while maintaining ≥ +2°C dew-point margin.	1,460 MWh/yr +1,450 econ hrs	**$167,900 / yr**
3. TOU Arbitrage & UPS BESS Shaving (Modules #11, #12)	Pre-cools thermal mass before 16:00–21:00 peak tariff & shaves 420 kW utility billing peaks via UPS Li-Ion BESS.	420 kW shaved + TOU shift	**$124,300 / yr**
4. Predictive CBM & SLA Defense (Modules #7, #14, #15, #16)	Eliminates premature calendar filter swaps, emergency strainer callouts, NOC overtime & disputed tenant SLA credits.	-90% alarm noise -35% filter spend	**$95,600 / yr**
TOTAL DIRECT RECURRING OPEX BENEFIT (PER 5.0 MW DATA HALL)		**4,175 MWh / yr**	**$700,000 / yr**

PUE & Sustainability Impact (5 MW Hall)



ANNUAL ESG & WATER REDUCTION (CSRD AUDIT-READY)
1,380 Metric Tons CO_2e Avoided • 4.8 Million Liters Tower Water Saved

The "Hidden Multiplier": Reclaimed Sellable Capacity

By balancing 3-phase PDU whips (`PRD-08`), lowering peak cooling parasitic kW, and proving N–1 summer ride-through (`PRD-02/03`), operators typically unlock **~650 kW (13%) of stranded IT capacity** in an existing 5 MW hall:

- **Avoided Tier-III New-Build CapEx** (650 kW × $8,500 / kW industry benchmark) — **$5.52M**
- **OR New AI Colocation Lease Revenue** (650 kW × $150 / kW-mo wholesale AI rate) — **+$1.17M/yr**

Campus-Scale Economics Summary (20 MW Colocation / Enterprise AI Campus = 4 × 5 MW Halls)

Across a 20 MW facility footprint, deploying Pandita's Aether-DCTWIN delivers **$2.80M / year in direct recurring energy & maintenance OpEx savings**, unlocks **2.6 MW of sellable AI rack capacity** (worth **$4.68M / year** in incremental wholesale lease revenue or **$22.1M** in deferred Tier-III construction CapEx), and achieves full software payback in **under 5 months**.

Winning Every Seat at the Table: Stakeholder Value Alignment

Enterprise data center purchases require sign-off from four distinct leaders. Aether-DCTWIN turns historical blockers into champions:

VP OF CRITICAL FACILITIES / CHIEF ENGINEER

Zero-Outage Sleep-at-Night Confidence

- Continuous 30-second Tier-III N–1 ride-through clock catches thermal margin erosion before an outage occurs.
- Dew-point guard prevents CDU manifold condensation; BACnet Priority-8 watchdog guarantees instant local PID fallback.
- Collapses 50+ alarm floods into 1 root-cause CMMS work order with auto-generated ASHRAE LOTO checklists.

SVP COLOCATION SALES & CAPACITY PLANNING

Close High-Density AI Deals Faster

- Self-service Fit-Out Planner tests 500 kW+ liquid/air deals under summer design-day wet-bulb in 60 seconds.
- Reclaims 10–20% stranded sellable kW across unbalanced 3-phase row PDUs without waiting on utility upgrades.
- Automated tenant ESG/WUE certificates & demarcation SLA ledgers differentiate RFPs for hyperscale AI tenants.

CFO & ENERGY / SUSTAINABILITY OFFICER

Auditable OpEx Cut & Deferred CapEx

- Shadow-Mode proves exact kWh and demand-charge savings empirically before full commercial expansion.
- Monetizes existing UPS Li-Ion batteries via peak-demand shaving & Time-of-Use tariff arbitrage without degrading backup reserve.
- Delivers board-ready 5-year NPV reports and CSRD-ready Scope-2 carbon & evaporative water accounting.

CISO & OT / BMS NETWORK SECURITY

Air-Gapped & 100% Agentless

- Deploys as a self-contained on-premises Docker / Kubernetes appliance inside the facility OT/BMS VLAN.
- **Zero** agents, credentials, or network routes into customer IT racks, hypervisors, or GPU fabrics.
- Starts 100% read-only (`Stage 1 Shadow Mode`) with hardware-clamped register whitelists for Stage 3.

Turnkey 30-Day Zero-Risk Pilot Roadmap

Customers validate Aether-DCTWIN on a live hall in 30 days with **zero downtime, zero hardware retrofits, and < 4 hours of facility engineering time**:

PHASE 1 • DAYS 1 – 3

BYOD Ingest & Calibration

- Load site topology into declarative YAML blueprint.
- Ingest 7–30 days of BMS/DCIM CSV or PI history + connect read-only BACnet/SNMP tap.
- Online RLS calibrates zone physics to <0.4°C RMSE.

PHASE 2 • DAYS 4 – 21

Read-Only Shadow Proof

- Run live Shadow Validator alongside legacy BMS PID.
- Deliver immediate quick wins: 3-Phase Stranded kW Audit, Tier-III N–1 Ride-Through Report & CBM RUL scan.
- Accumulate verified $ & kWh savings scoreboard.

PHASE 3 • DAYS 22 – 30

Advisory & ROI Sign-Off

- Enable NOC Advisory Queue & test BACnet Priority-8 watchdog fail-safe on 1 pilot zone.
- Verify live kW drop on facility main meter.
- Export CFO Executive ROI Report for campus rollout.

Deployment & Integration Specifications at a Glance

Supported Facility Protocols: BACnet/IP (Priority Array 1–16), Modbus TCP (CoolIT, Motivair, Vertiv CDUs/CRACs), SNMPv3 EPMS (Schneider PowerLogic, Eaton, ServerTech/Raritan PDUs), MQTT Sparkplug B, AVEVA/OSIsoft PI & CSV/Parquet BYOD.

Hosting & Enterprise IT/OT Fit: Single-Container Docker / Kubernetes On-Premises OT Appliance (air-gap capable) or Private Cloud VPC • Embedded DuckDB + Parquet Historian • ServiceNow & IBM Maximo Webhook Ready.